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Warped Time and Space, and Tidal Gravity



Einstein's Law of Time Warps

Einstein struggled to understand gravity on and off from 1907 onward. Finally in 1912 he had a brilliant inspiration. Time, he realized, must be warped by the masses of heavy bodies such as the Earth or a black hole, and that warping is responsible for gravity. He embodied this insight in what I like to call "Einstein's law of time warps," a precise mathematical formula[8] that I describe qualitatively this way: *Everything likes to live where it will age the most slowly, and gravity pulls it there*.

The greater the slowing of time, the stronger gravity's pull. On Earth, where time is slowed by only a few microseconds per day, gravity's pull is modest. On the surface of a neutron star, where time is slowed by a few hours per day, gravity's pull is enormous. At the surface of a black hole, time is slowed to a halt, whence gravity's pull is so humungous that nothing can escape, not even light.